

Neel Tamtam

Systems engineer with depth across GPU, embedded, and systems languages

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[github](#) | [linkedin](#) | [website](#)

SKILLS & CERTIFICATIONS

Certifications	CUDA Accelerated Computing JLPT N1
Languages	C Rust C++ Zig ARM Assembly RISC-V Assembly Python
Embedded	FreeRTOS STM32 ESP32 I2C I2S UART
Graphics	CUDA Vulkan OpenGL GLSL HLSL
Tooling	GDB OpenOCD Probe-rs J-Link ST-Link
Network	AWS GCP PostgreSQL MongoDB TCP/IP
AI/ML	PyTorch vLLM ONNX

EXPERIENCE

National Institutes of Health

July 2024 - Present

Software Engineer, National Library of Medicine

Remote

- Deployed vLLM continuous batching on GPU, cutting 600K FDA record structuring from 7+ days to 47 hours.
- Diagnosed Oracle Linux server crash via kernel dump analysis; resolved OOM by chunking XML batch pipeline.
- Migrated FDA-to-NLM batch pipeline from serial bash cronjobs to parallel Airflow DAGs, cutting runtime by 38%.
- Owned anomaly detection system from MVP to internal board presentation, securing over \$1M in NLM funding.

NOTABLE PROJECTS

CUDA-Accelerated Ray Tracing Simulator

[GitHub](#)

Graphics Programming, GPU Computing

C++20, CUDA, Vulkan, ImGui

- Architected a CUDA path-tracing engine with a bare-metal Vulkan rasterization pipeline replacing OpenGL.
- Optimized warp utilization 5.2x by refactoring to a wavefront architecture with per-bounce material sorting.
- Implemented linear BVH on GPU, reducing intersection complexity from $O(n)$ to $O(\log n)$ for complex scenes.
- Engineered zero-copy CUDA-Vulkan interop, exporting traced frames to VkImages with ACES tonemapping.

Custom Real-Time Operating System

[GitHub](#)

Embedded Systems (STM32F407)

C, Rust, ARM Assembly, GDB, J-Link, ST-Link, I2C, I2S

- Designed preemptive kernel with priority round-robin scheduling, validated with a Rust FFI traffic simulator.
- Achieved sub-microsecond scheduling jitter (70ns) and 100% determinism for intensive RTOS workloads.
- Implemented assembly context switching; optimized state preservation to $<4\mu s$ for 32+ concurrent tasks.
- Built inter-task sync primitives with blocking/non-blocking semantics for custom CS43L22, I2C, and I2S drivers.

Multiplayer Japanese Kanji Trainer in Rust

[Live Site](#) | [GitHub](#)

High-Performance Web

Rust, Tokio, Axum, Leptos, PostgreSQL

- Built and deployed a multiplayer kanji game in active beta for beginner (N5) to advanced (N1) Japanese learners.
- Engineered a scalable 100% Rust full-stack architecture using Leptos (WASM), Axum, Tokio, and PostgreSQL.
- Implemented JWT auth with Argon2 hashing, rate limiting via tower-governor, and SQLx compile-time queries.
- Orchestrated low-latency, real-time game state synchronization and interactive lobby chat via WebSockets.

MERGED CONTRIBUTIONS

gfx-rs/wgpu

[PRs](#)

- Fixed incorrect atomic load/store emission in GLSL/HLSL backends using atomicOr/atomicExchange polyfills.
- Replaced explicit Vulkan swapchain destroy calls with Rust-native Drop impl for automatic RAII cleanup.

ziglang/zig

[PRs](#)

- Ported POSIX doubly-linked list functions insque/remque from musl/wasi-libc C to native Zig implementation.
- Added ABI register name aliases (ra, sp, a0-a7, etc.) as valid inline assembly clobbers for RISC-V targets.

EDUCATION

Bachelor of Computer Science and Japanese

Aug 2020 - May 2024

Austin College

Texas, USA

Semester Abroad - Japanese Language

Jan 2023 - May 2023

Osaka Daigakuin University

Osaka, Japan

Summer Abroad - Machine Learning

June 2023 - Aug 2023

Yonsei University

Seoul, South Korea